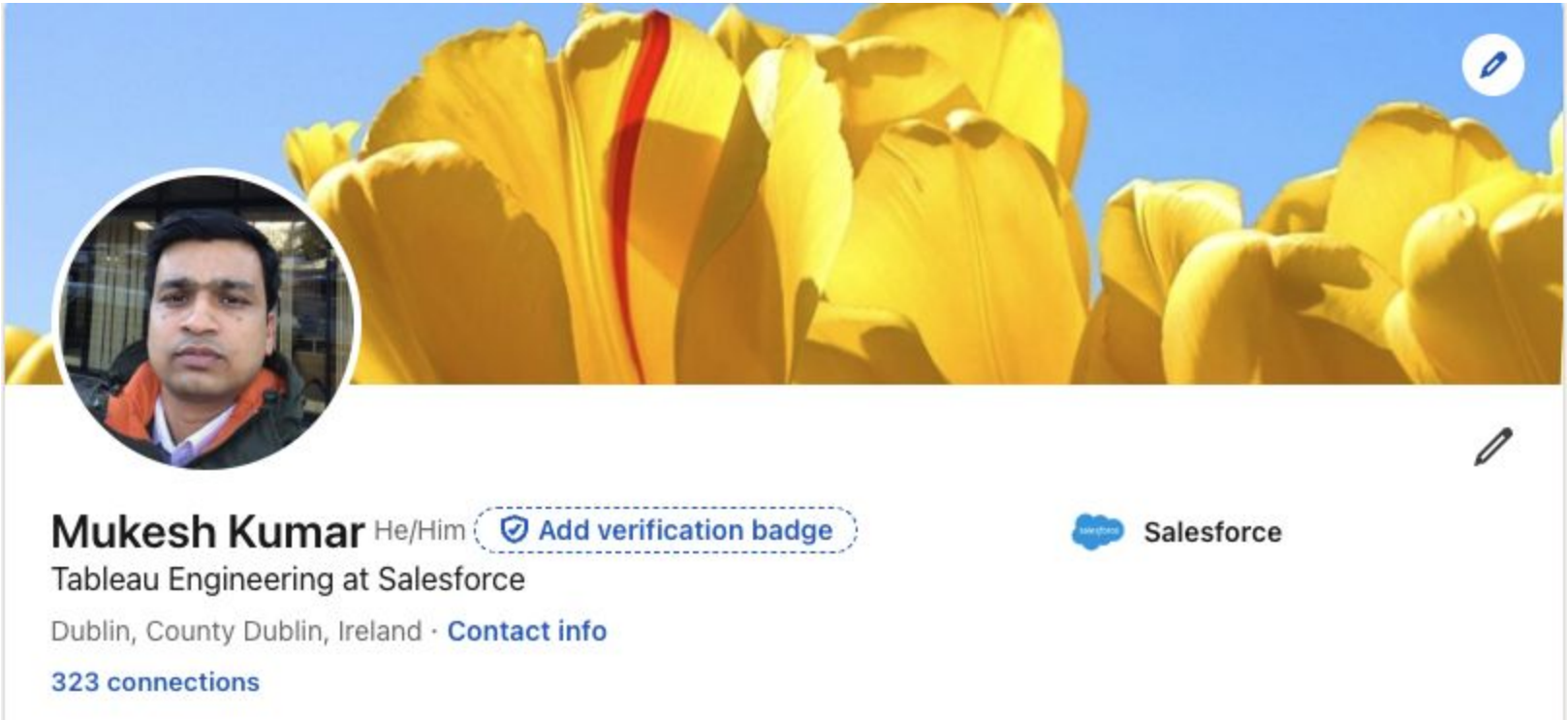
linkedin.com/in/mukesh-kumar-96082765



Visualization Science

Visualization Science is the science of seeing and understanding data.

Why do we call it the “science” of data visualization?

- Built on the proven observations of cognitive and perceptual psychology.

How do we perceive data?

- Different parts of our brain process the data and assign a meaning to different things like position, color and size.
- Allows us to form ideas and patterns rapidly.

Visualization Science

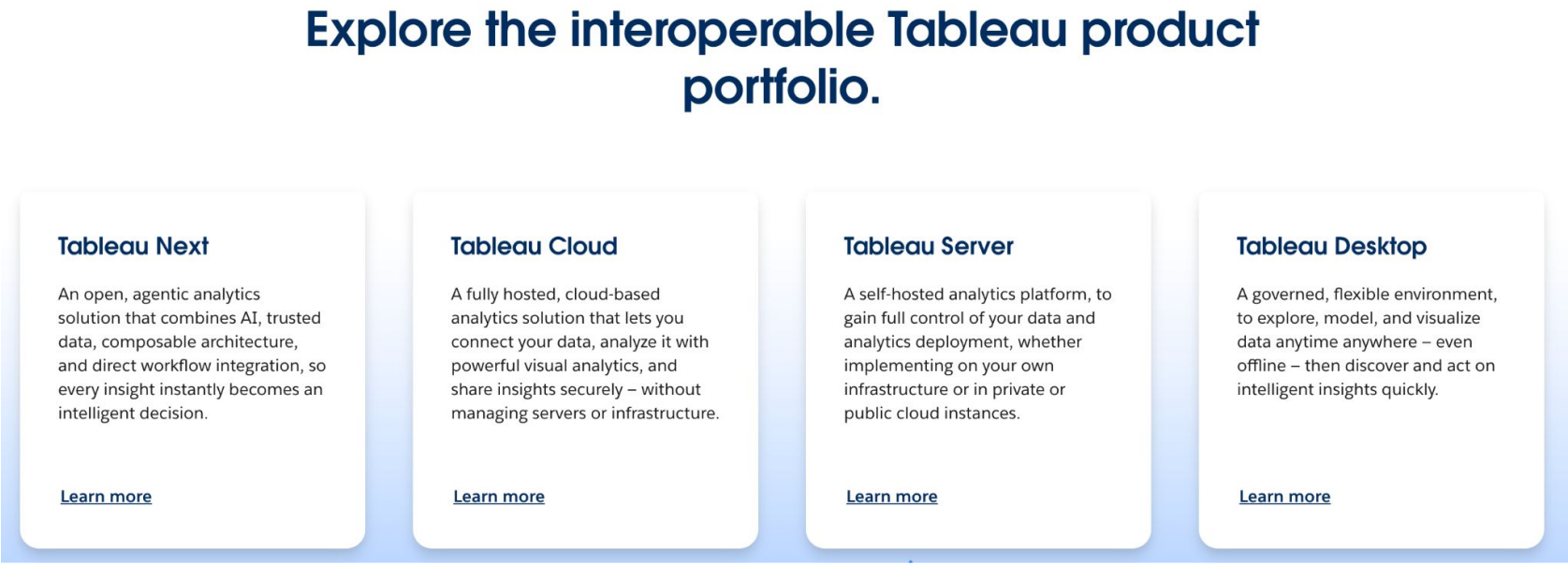


THINKING IS SLOW. SEEING IS FAST.

Tableau

Tableau is a powerful tool that turns raw data into **clear and actionable insights**. It does not only show numbers—it helps us **understand the story behind the data**.

Let’s get started and take a look at Tableau in action



Comparison guide

Features and Capabilities	Tableau Desktop Public Edition	Tableau Desktop Free Edition	Tableau Desktop Professional Edition
Privacy and security	✓	✓	✓
Save visualizations locally	✓	✓	✓
Autorecovery	✓	✓	✓
Publish or embed visualizations on other public sites	✗ * Embedding options available after publishing to Tableau Public	✗	✓
Publish visualizations to Tableau Public	✓	✗	✓
Publish visualizations to Tableau Cloud or Tableau Server	✗	✗	✓
Share packaged workbooks with other users	✗	✗	✓

https://help.tableau.com/current/pro/desktop/en-us/desktop_comparison.htm

Live data refresh	Limited data refresh (Google Drive only. Data is refreshed automatically once every 24 hours)	✓	✓
Data sources	Limited data source options (Google Drive, JSON files, Microsoft Excel 2007 or later, OData, PDF, Spatial files, Statistical files, Text files)	✓ * Published data sources on Tableau Cloud or Tableau Server are not supported	✓ (Can connect to all data sources)
Data source rows	✓ 15M rows	✓ Unlimited	✓ Unlimited
Cloud Storage	✓ 10GB on Tableau Public	✗	✓ Tableau Cloud and Tableau Server if applicable
Local Storage	✓	✓	✓

5 Critical Questions

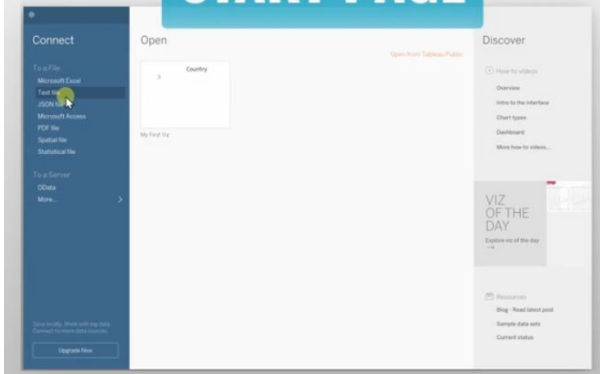
1. What story is your data trying to deliver?
2. Who will you present your results to?
3. How much data is there?
4. What data type are you working with?
5. How are these data elements related to each other?



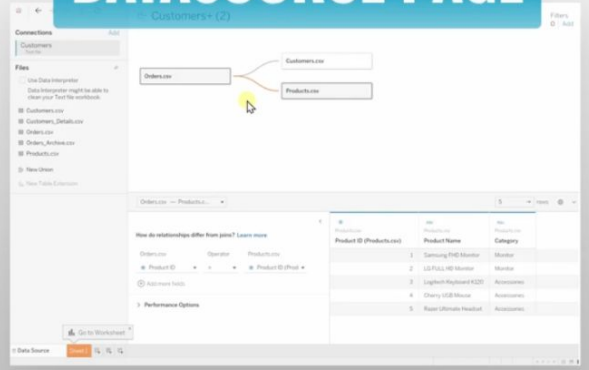
- Connect to and Prepare Data
- Build Charts and Analyze Data
- Create Dashboards
- Create Stories
- Format Worksheets and Workbooks
- Save Work
- Publish Data Sources and Workbooks
- Use Tableau on the Web

TABLEAU WORKBOOK

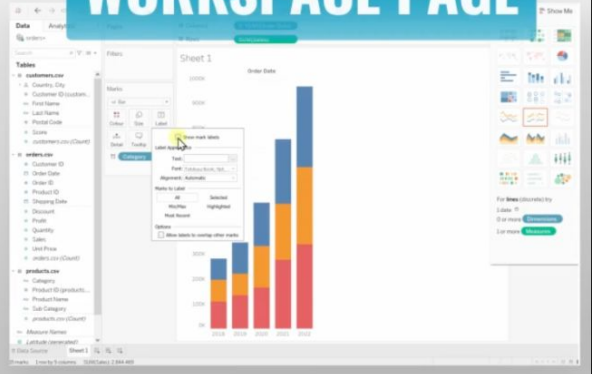
START PAGE



DATASOURCE PAGE



WORKSPACE PAGE



All fields in a data source have a data type. The data type reflects the kind of information stored in that field, for example integers (410), dates (1/23/2025) and strings ("Belgium").

Data type icons in Tableau

Icon	Data type
Abc	Text (string) values
	Date values
	Date & Time values
#	Numerical values (integer or decimal)
T|F	Boolean values (relational only)
	Geographic values (used with maps)
	Image role (used with image link URLs)
	Cluster Group (used with Find Clusters in Data)

You can change the data type for a field either on the **Data Source** page or in the **Data** pane.

Need more data?

Drag tables here to relate them. Learn more

7 fields 5 rows

5 → rows

Customers.csv Customer ID	Customers.csv First Name	Customers.csv Last Name	Customers.csv Postal Code	Customers.csv City
1	Maria	Cramer	12209	Berlin
2	John	Steel	98128	Seattle
3	Georg	Pipps	75016	Paris
4	Martin	Mueller	70563	Stuttgart
5	Peter	Franken	94117	San Francisc

ysical Table	Remote Field N...
ustomers.csv	Customer_ID
ustomers.csv	First_Name
ustomers.csv	Last_Name

DATA GRID

Converting a measure in the Data pane into a dimension

- When you first connect to a data source, Tableau assigns most fields that contain quantitative, numerical information (that is, fields where the values are numbers) as **measure fields** in the **Data** pane.

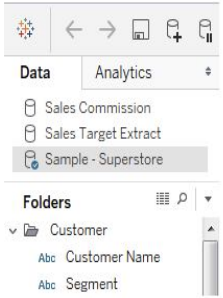
- The **exception is for fields** where the name suggests the **data type, such as Year or Month (**which Tableau would **identify as Date dimensions**) or fields containing words like “ID” and “Key,” which Tableau would categorize as dimensions, even when they are numeric.

- However, **you might decide that some of these fields that Tableau has categorized as measures should actually be dimensions.** Postal codes are a classic example—they often consist entirely of numbers, but the information is categorical and not continuous—**you would never want to aggregate postal codes by adding or averaging them**.

- Similarly, **a field containing individuals′ ages may be categorized as a measure by default in Tableau because it contains numeric data.** In some cases **you may want to add or average ages**, but **you might also want to look at each individual age as a bin or category, i**n which case you want Tableau to create headers for this field rather than an axis. **If this is how you want to use age in your view, you can convert the field to a dimension.**

Perform common tasks in the Data pane

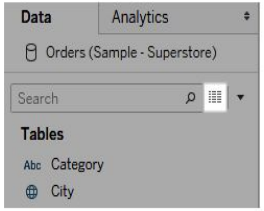
To select a data source connection for analysis, click the data source connection name in the Data pane. For more details, see Navigate Data Sources in the Data Pane.



To view a context menu for the data source, click **Data** in the top menu and then select the data source in the list. For more information, see Edit Data Sources.

To search for fields in the Data pane, click the magnifying class icon and then type in the text box. For more details, see Find fields.

To see the underlying data, click the **View Data** icon at the top of the **Data** pane.



In cases where Tableau has misclassified a field as a dimension or a measure, you can convert it and change its role.

To convert a measure to a dimension in the Data pane, do either of the following.

- **Click and drag the field from a measures area in the Data pane** and drop it into a dimensions area

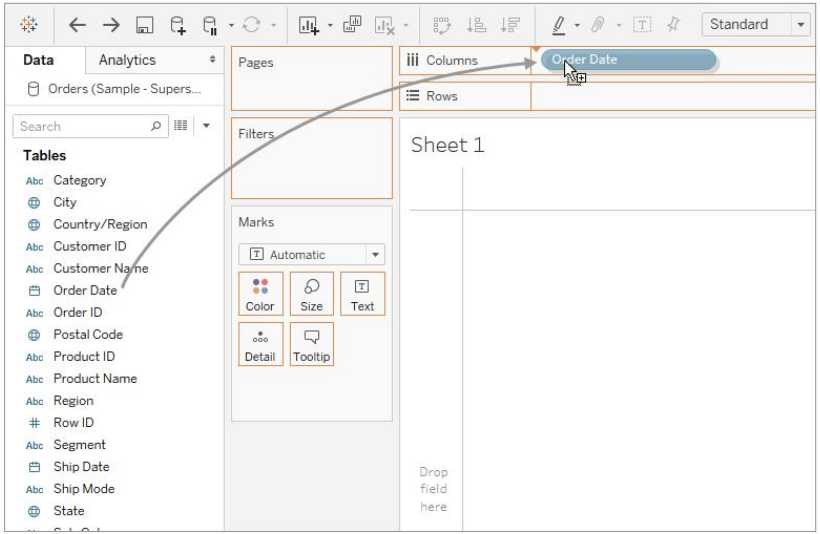
- Right-click (control-click) **the measure in the Data pane** and select **Convert to Dimension**.

Build a view from scratch

These steps show how to build a basic view that shows year-by-year profit.

1. From the **Dimensions** area in the **Data** pane, drag the **Order Date** field to the **Columns** shelf. You might need to expand the Order hierarchy to see Order Date.

   When you drag a field over a shelf, a plus sign indicates that the shelf can accept the field.

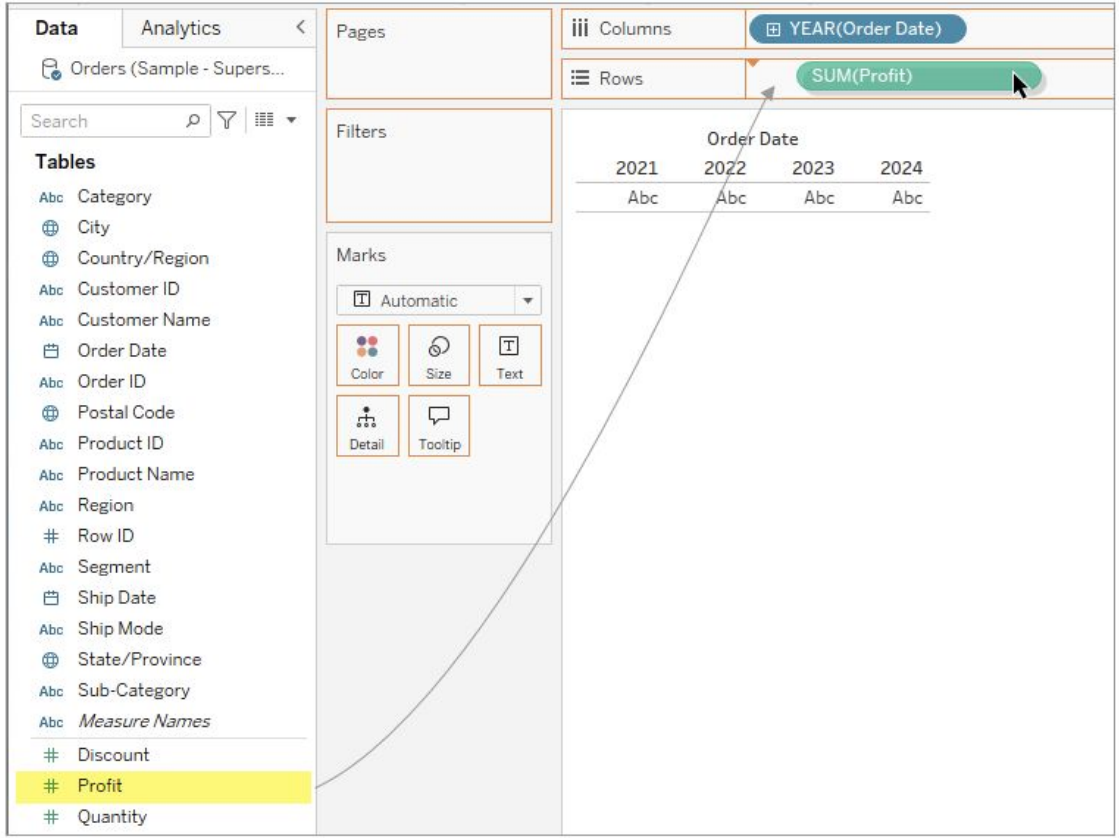


The resulting table has four columns and one row.

Each column header represents a member of the Order Date field (the default date level is YEAR).

Each cell contains a placeholder "Abc" label, which indicates that the current mark type for this view is text.

2. From the **Data** pane, drag the **Profit** field to the **Rows** shelf.



In Tableau, discrete data (blue pills) represents distinct, separate categories or counts, creating headers, while continuous data (**green pills**) represents an unbroken range of numerical values, creating axes. Discrete data (e.g., regions, product names) classifies data, whereas continuous data (e.g., sales, profit) measure

The **Profit** field is colored green on the **Rows** shelf, indicating that it's continuous.

Also, the field name changes to **SUM(Profit)**. That's because measures are automatically aggregated when we add them to the view, and the default aggregation for this measure is SUM.

Tableau transforms the table into a line chart and creates a vertical axis (along the left side) for the measure.

A line chart is a great way to compare data over time and identify trends effectively.

This line chart shows profit over time. Each point along the line shows the sum of profit for the corresponding year.



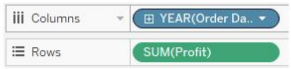
The next step is to drill a little deeper into the time dimension.

Drill into the data

This step shows how you can modify the view to show quarters in addition to years. As you drill down into the hierarchy, the view changes to become a nested table.

You can show **[Order Date]** by quarters using either of the following methods:

- Select the plus button on the left side of the field **YEAR(Order Date)** field on Columns.



- Drag the **Order Date** field (again) from the Data pane and drop it on the **Columns** shelf to the right of **Year(Order Date)**.

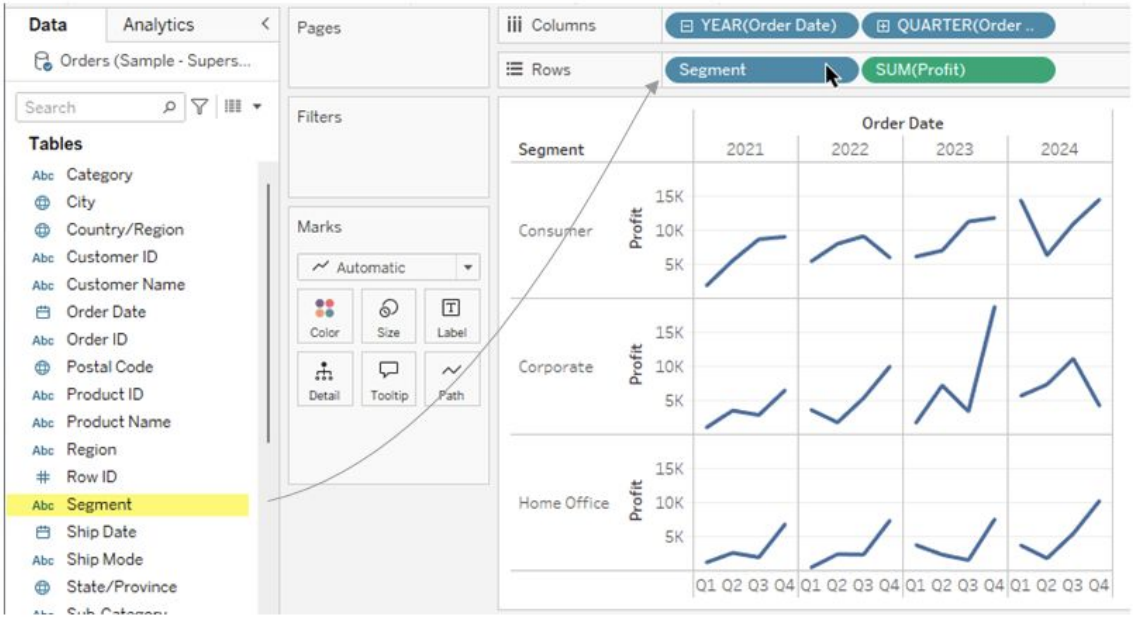
The new dimension divides the view into separate panes for each year. Each pane, in turn, has columns for quarters. This view is called a nested table because it displays multiple headers, with quarters nested within years. The word "headers" is a little misleading in this example because the year headers are displayed at the top of the chart, while the quarter headers are displayed at the bottom.



Increase the level of detail with small multiples

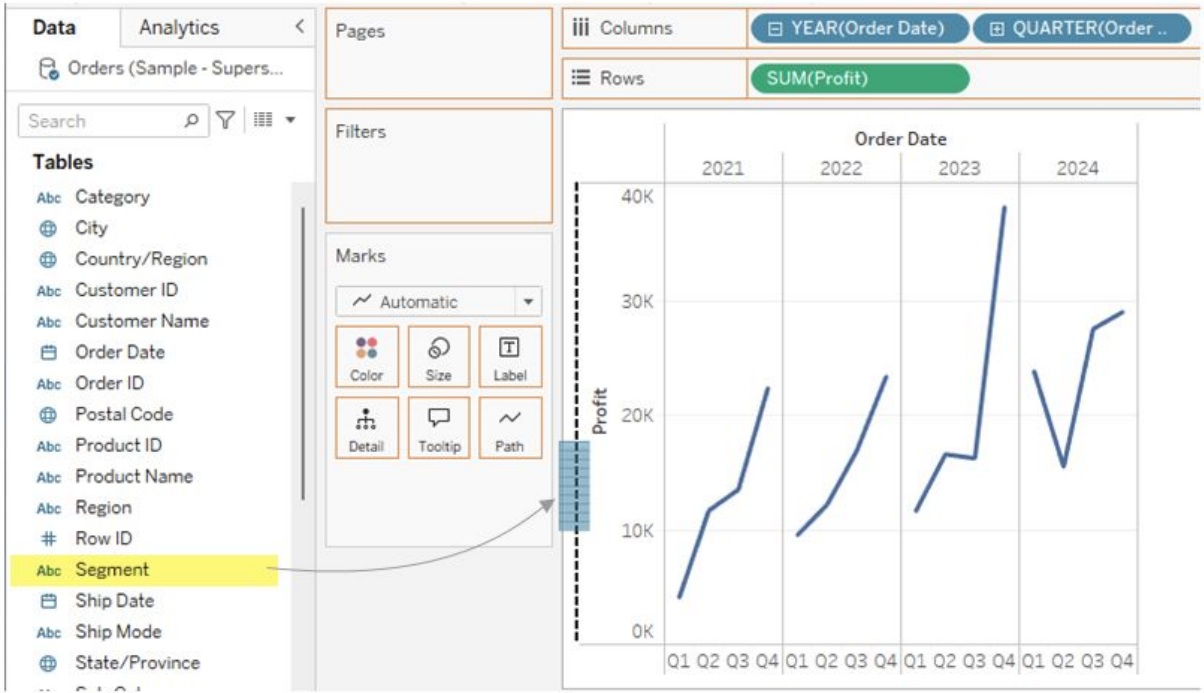
This step shows how you can modify the nested table view to add customer segment. It creates what we call a *small multiples* view.

Drag the **Segment** dimension from the **Data** pane, and then drop it just to the left of **SUM(Profit)** on Rows.



The field is added to the Rows shelf and row headers are created. Each header represents a member of the Segment field.

Optionally, you can achieve the same result by dropping **Segment** just to the left of the Profit axis in the view (shown in the image below). Tableau often supports multiple ways to add fields to the view.



Note: Tableau doesn't allow you to place a dimension to the right of a measure on either the Rows or Columns shelves because that visual structure doesn't make sense in the view.

The new dimension divides the view into 12 panes, one for each combination of year and segment. This view is a more complex example of a nested table. Any view that contains this sort of grid of individual charts is referred to as a small multiples view.



This could be the right amount of granularity for the view. Another option at this point would be to filter some of the data out of the view to simplify it.

Filter the view to focus exploration

The view updates to only show data rows where Order Date is 2023 or 2024. Tableau can now allocate more space to the data that interests you.



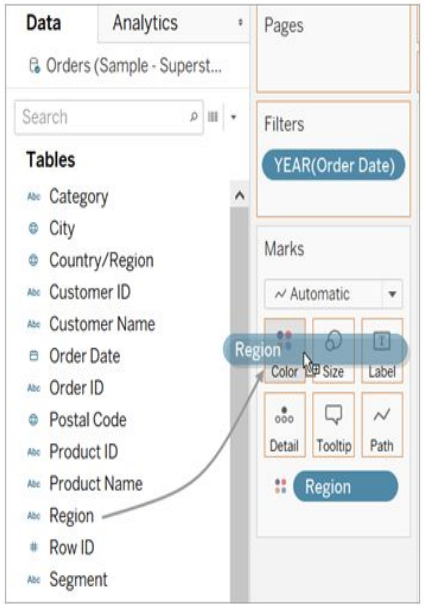
Next, you'll increase the detail in your data exploration by dragging a field to the Color property on the Marks card.

Use the Marks card to add depth to your analysis

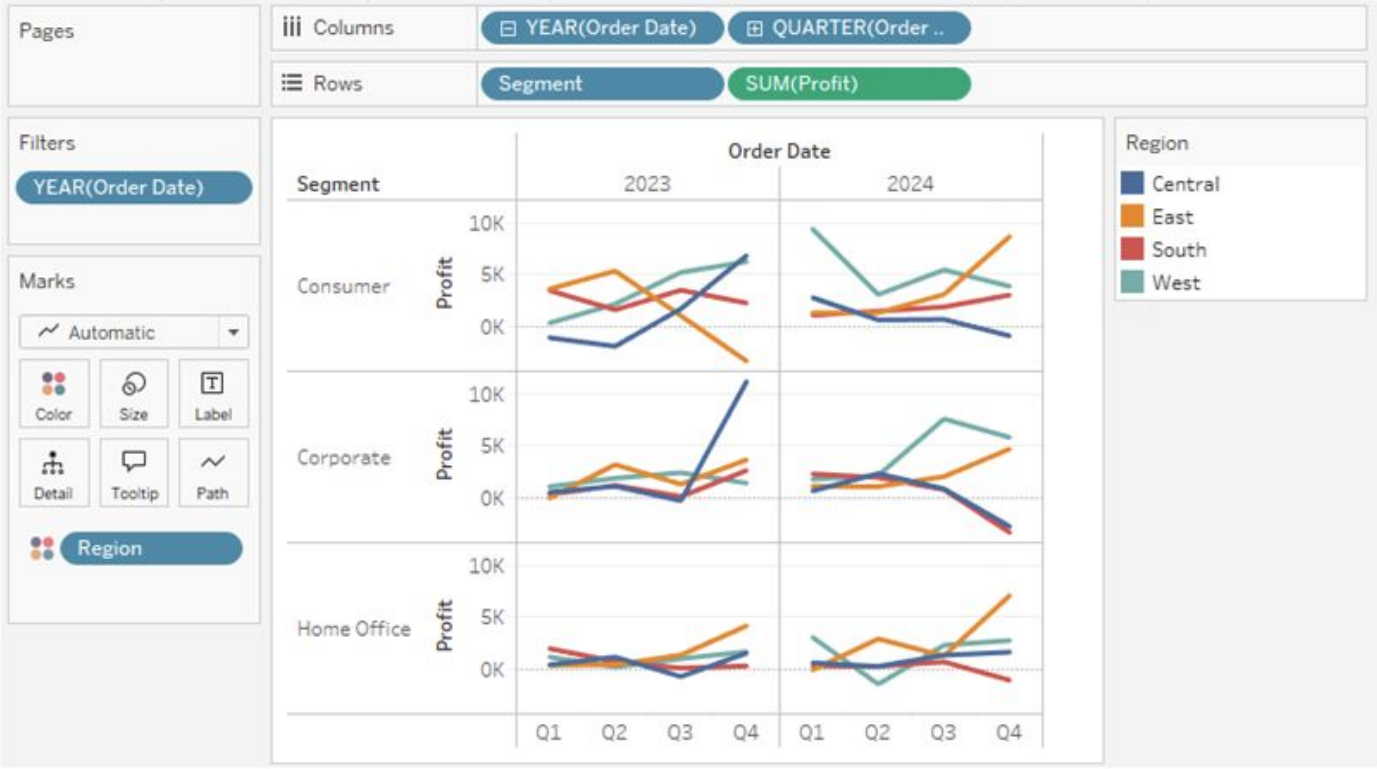
This step shows how you can modify the view to color the marks based on the region.

1. Drag the **Region** dimension from the **Data** pane and drop it on **Color**.

   Placing a dimension on **Color** separates the marks according to the members in the dimension, and assigns a unique color to each member. The color legend displays each member name and its associated color.



The view now shows profit for each customer segment and region for 2018 and 2019.



Each pane now has four lines, one for each region.

The viz is now showing profit data summarized the Region level of detail.

- Fitlers
- Parameters
- Calculated field

  etc….

Simple Summary

- **Filters** → Control *what data you see*
- **Parameters** → Let users *control inputs*
- **Calculated Fields** → Create *new insights from data*

Key Concept Summary

- **Filters** → *View-specific, reduce data, tied to fields*
- **Parameters** → *Workbook-wide, independent, require logic to work*
- **Calculated Fields** → *Create new fields, can act as dimensions or measures*

🔷 Key Differences

Feature	Filters	Parameters	Calculated Fields
Purpose	Limit data	User input control	Create new data/metrics
User Interaction	Moderate	High	Indirect
Data Impact	Reduces dataset	Doesn’t change data	Transforms data
Flexibility	Medium	Very High	Very High

Feature	Filters	Parameters	Calculated Fields
Scope	View / Worksheet / Dashboard	**Workbook-wide**	Workbook or Data Source
Tied to Data	Yes (Dimensions & Measures)	❌ No	✅ Yes
Data Reduction	✅ Yes	❌ No	❌ No (transforms instead)
User Input	Limited	✅ High	Indirect
Requires Other	❌ No	✅ Needs Calculated Field	❌ No
Storage	Workbook	Workbook	Workbook / Data Source

🔷 1. Filters (Tableau)

Definition:

Filters restrict the data displayed in a worksheet or dashboard.

Key Points:

- ✅ **Primarily View-specific** (applied to a worksheet, but can be applied to multiple views)
- ✅ Can be applied at:
  - Data Source level
  - Worksheet level
  - Dashboard level
- ✅ **Act directly on Dimensions & Measures**
- ✅ Reduce the volume of data being processed/displayed
- ✅ Stored within the **workbook**, not the original data source

Special Notes:

- Context filters improve performance by creating a subset
- Filter order affects results (important in complex dashboards)

Types of Filters:

- **Dimension Filters** – filter categorical data (e.g., Region, Category)
- **Measure Filters** – filter numeric values (e.g., Sales > 1000)
- **Date Filters** – filter based on time (e.g., last 3 months)
- **Context Filters** – applied first; improve performance
- **Top/Condition Filters** – show top N values or based on rules

Example:

Show only sales where *Region = Europe*

Use Case:

Focus analysis on relevant subsets of data

🔷 2. Parameters (Tableau)

Definition:

Parameters are user-controlled input values that can dynamically change calculations and views.

Key Points:

- ✅ **Workbook-specific (global across sheets)**
- ❌ **Not tied to Dimensions or Measures directly**
- ❌ Do NOT filter data by themselves
- ✅ Require **Calculated Fields** or filters to have an effect
- ✅ Accept only **one value at a time**
- ✅ Can be shown as controls (dropdown, slider, etc.)

Special Notes:

- Independent of the data source
- Can simulate filters, metric switching, or what-if analysis

Example:

User selects a value (e.g., Top 10 customers)

Use Case:

Create interactive dashboards (e.g., switch between metrics like Profit/Sales)

🔷 3. Calculated Fields (Tableau)

Definition:

Custom fields created using formulas to transform or derive new data.

Key Points:

- ✅ Can be:
  - **Workbook-specific** OR
  - **Data source-specific** (if created at data source level)
- ✅ Become part of **Dimensions or Measures** depending on formula
- ✅ Reusable across worksheets
- ✅ Can use:
  - Fields
  - Parameters
  - Functions (logical, math, date, etc.)

Special Notes:

- Can drive filters, labels, tooltips, and aggregations
- Can be row-level or aggregate-level

Common Functions:

- Arithmetic: `SUM(Sales) – SUM(Cost)`
- Logical: `IF`, `CASE`
- String: `LEFT`, `CONCAT`
- Date: `DATEDIFF`, `TODAY`

Example:

```
Profit Ratio = SUM(Profit) / SUM(Sales)
```

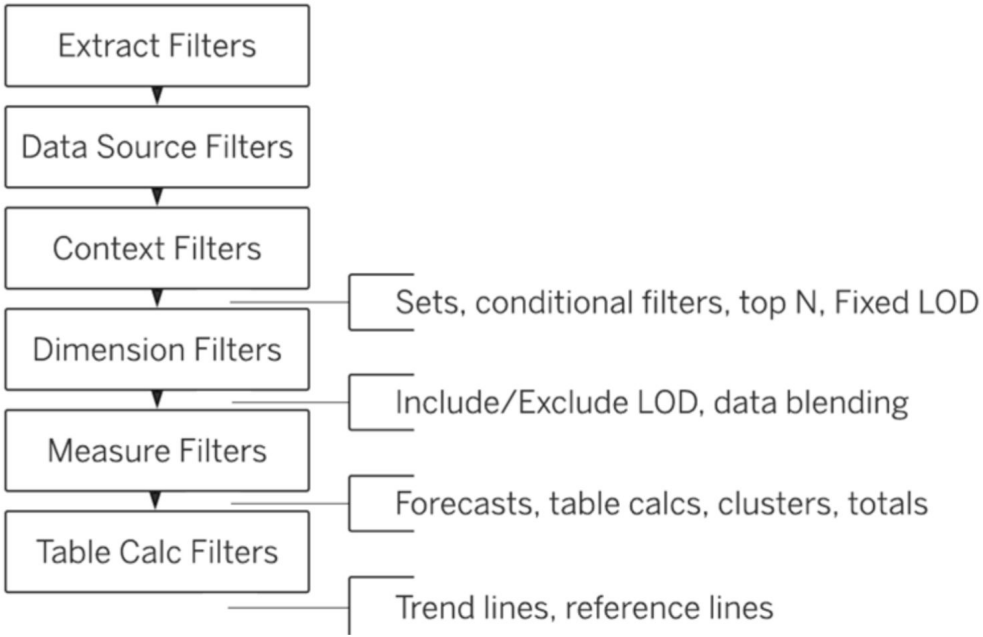
Use Case:

Create new insights or metrics not directly available in the dataset

About the order of operations (aka query pipeline)

Sometimes, you might expect Tableau to execute filters in one order, but the order of operations dictates that they be executed in a different order, which gives you unexpected results. When this happens, you can sometimes change the order in which operations are executed in the pipeline.

The Tableau order of operations includes the following, from top to bottom.



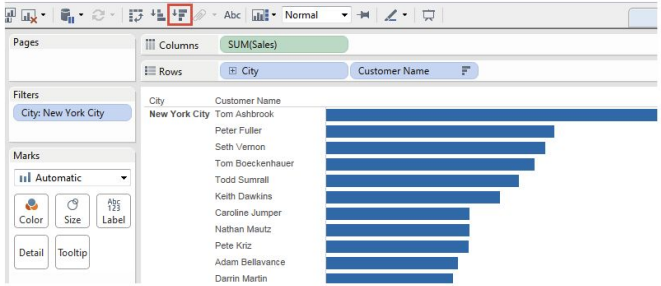
https://help.tableau.com/current/pro/desktop/en-us/order_of_operations.htm

Convert a Dimension Filter to a Context Filter

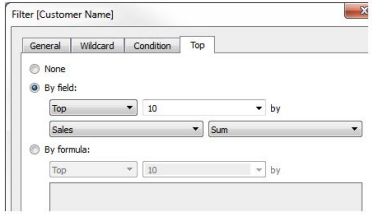
Here are the steps for building this view.

1. Drag **Sales** to **Columns**.
2. Drag **City** and **[Customer Name]** to **Rows**.
3. Drag **City** from the **Data** pane again, this time to **Filters**. On the **General** tab in the Filter dialog box, set the filter to show just a single value: New York City. Do this by clicking **None** and then choosing **New York City**.

   This creates a general dimension filter.
4. Click the **Sort Descending** button ( ) on the toolbar. Your view now looks like this:



   Note the first few names in the list: Ashbrook, Fuller, Vernon, etc.
5. Now drag **[Customer Name]** from the **Data** pane to **Filters**, and create a Top 10 Filter, to show only the top 10 customers by total sales:



   After you apply this second filter, the view looks right, but notice that the names shown are no longer the same as

In this example, **the view addresses the following question: Who are the top 10 customers, by total sales, in New York City?**

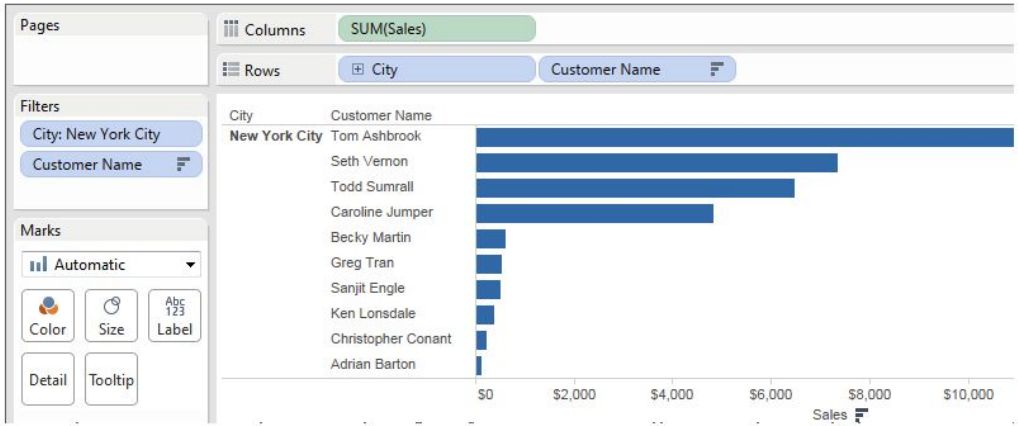
The view contains two dimension filters, one that you create on the **General** tab in the Filters dialog box, and the other on the **Top N** tab.

The problem is that these filters are executing simultaneously, whereas you would like to general filter to be applied before the top n filter, so that the top n filter can act on the results as previously filtered by the general filter.

The solution is to redefine one of the filters as a context filter so that a clear order of precedence is established.

Convert a Dimension Filter to a Context Filter

After you apply this second filter, the view looks right, but notice that the names shown are no longer the same as before:



What happened to Peter Fuller, formerly in second place? The goal was to show the top 10 customers in New York City, but now the view is actually showing the top 10 customers overall.

The problem is that top and general dimension filters are applied simultaneously—they are both dimension filters, and they appear in the same place in the Tableau order of operations:

In this example, **the view addresses the following question: Who are the top 10 customers, by total sales, in New York City?**

The view contains two dimension filters, one that you create on the **General** tab in the Filters dialog box, and the other on the **Top N** tab.

Tableau Task

Create a Dynamic Measure to switch between 3 measures :
Sales, profit and quantity in the same view

Download Tableau desktop public Edition

https://public.tableau.com/

Trending Viz

https://www.tableau.com/data-insights/dashboard-showcase/sales-commission-model

Tableau Dashboard Showcase Examples

https://www.tableau.com/data-insights/dashboard-showcase

Which chart or graph is right for you?

https://www.tableau.com/sites/default/files/2022-03/which_chart_or_graph_is_right_for_youwp.pdf

Tableau Learn

https://public.tableau.com/app/learn/how-to-videos

Tableau for Students

https://www.tableau.com/academic/students

Watch a video:

You can see many Tableau concepts and product features discussed and demonstrated on the Tableau Tim website

Udemy Example:

https://public.tableau.com/app/profile/maureen.okonkwo/viz/UdemyCourseAnalysisDashboard/Dashboard

CV
https://public.tableau.com/app/profile/michelle.frayman/viz/ResViz2021/VisualResume

Carbon Clock
https://public.tableau.com/app/profile/harim.jung/viz/GlobalCODashboardPoweredbyTableauAgent/TableauAgentClimateDashboard

Q&A

Thank you !